

Assignment 7: CNN and Image Classification

1. What does image classification mean? Explain with one example.

Image classification means teaching a computer to look at an image and decide which category it belongs to.

For example, if we give the model a picture of a shirt, the model should classify it as “shirt.” If we give it a picture of a sneaker, it should classify it as “sneaker.”

2. Why is using a Dense Network not always a suitable choice for images?

A Dense Network is not always suitable for images because it treats every pixel as a separate input. It does not naturally understand the position of pixels or the shape of objects in the image.

Images have patterns, such as edges, corners, lines, and textures. A Dense Network may miss these spatial patterns because it does not look at nearby pixels as connected parts of the image.

CNNs are better for images because they can learn local patterns from small areas of the image.

3. In a CNN, what is convolution, and what does a filter or kernel do?

Convolution is the process of moving a small filter across an image to detect useful patterns.

A filter, also called a kernel, is a small grid of numbers. It scans different parts of the image and looks for patterns such as edges, curves, textures, or shapes.

For example, one filter may learn to detect vertical lines, while another filter may learn to detect corners.

4. What is a feature map in a convolutional neural network?

A feature map is the output created after a filter scans the image.

It shows where a certain pattern was found in the image. If a filter is looking for edges, the feature map will highlight the parts of the image where edges appear.

In simple words, a feature map shows what the CNN noticed in the image.

5. Why is weight sharing used in CNNs? Explain.

Weight sharing means the same filter is reused across different parts of the image.

This is useful because the same pattern can appear anywhere in an image. For example, an edge could appear at the top, bottom, left, or right side of an image.

Weight sharing also reduces the number of parameters in the model. This makes the CNN more efficient and easier to train.

6. How does the ReLU activation function work? Write the ReLU output for the following values: [-3, 0, 2, 5]

ReLU stands for Rectified Linear Unit.

It changes all negative values to 0 and keeps positive values the same.

The input values are:

[-3, 0, 2, 5]

The ReLU output is:

[0, 0, 2, 5]

This happens because -3 becomes 0, 0 stays 0, 2 stays 2, and 5 stays 5.

7. What is max pooling, and why is it used in CNNs?

Max pooling reduces the size of the feature map by keeping the largest value from a small area.

For example, if a small area has the values 1, 3, 2, and 5, max pooling keeps 5.

Max pooling is used because it keeps the strongest feature and reduces unnecessary information. It also makes the model faster and helps the CNN focus on the most important patterns.

8. In the Fashion-MNIST code, why are images reshaped from 28×28 to $28 \times 28 \times 1$?

Fashion-MNIST images are grayscale images with a size of 28×28 pixels.

CNN models usually expect image data to include a channel dimension. For grayscale images, there is only one channel.

That is why the image is reshaped from 28×28 to $28 \times 28 \times 1$.

The “1” means the image has one color channel.

9. What is the difference between training data, validation data, and test data?

Training data is used to teach the model. The model learns patterns from this data.

Validation data is used during model development. It helps check how well the model is doing while we adjust the model.

Test data is used after training. It checks how well the final model performs on new data that it has not learned from.

10. What is overfitting? If training accuracy is high but validation accuracy is low, what does this indicate?

Overfitting means the model learns the training data too closely.

When this happens, the model performs very well on the training data, but it does not perform well on new data.

If training accuracy is high but validation accuracy is low, it means the model may be overfitting. The model is memorizing the training examples instead of learning general patterns that work on new images.